

existing industries as well for such purpose as personal transportation, air taxis or passenger drones, goods delivery, and inspections. Hybrid drone technology will revolutionise delivery for e-commerce companies like Amazon, Alibaba, eBay, etc.

Hybrid drones are designed with longer endurance and range for outdoor use. For example, Foxtech GAIA 160HY hybrid drone is a hexacopter equipped with a gasoline-electric hybrid power system. It has an onboard 2000W generator that drives high-efficiency motors to ensure a long flight time of about three hours. Besides the internal combustion engine, a 12S lipo battery package is installed onboard to offer emergency back-up power.

The exact scope and future of passenger drones remain uncertain since this technology is new. Advancement in aerial drone technology could result in a rapid proliferation of passenger drones for civilian use like air-taxis and air-ambulance services. Manufacturers of passenger hybrid drones are working to overcome many challenges including noise pollution, short flight times, airspace regulations, safety, and general operations. Some of the popular manufacturers of passenger hybrid drones include Ehang, Volocopter, Workhorse and Top Flight Technologies.

Ehang. The Ehang 184 was one of the first hybrid drones built for human transportation. Already tested extensively with human passengers, it is one of the most compact passenger drones.

Volocopter. The Volocopter from Germany probably looks like a tradi-



Fig. 3: Volocopter hybrid drone (Credit: www.volocopter.com/en)

tional helicopter. It is one of the few drones to use fibre-optic cables for its communication networks.

Workhorse Surefly. It was introduced in 2017 and uses both a gasoline generator and electric battery packs as power backups.

Top Flight Technologies. Their Airborg H6 1500 drone has good endurance capabilities and an advanced payload, providing an approximate flying time of over two hours. This drone can reach a maximum speed of 64km per hour while supporting a maximum payload of about 9kg. It can cover a range of 160km, withstanding unfavorable wind conditions up to 56km per hour. It has an advanced autopilot system responsible for calibration and tuning. This drone also consists of a 5000W rated hybrid engine, a tank with a 11.3-litre capacity, and a 16,000mAh LiPo battery.

What is an edible drone?

Don't be surprised when you hear that someone is eating a drone! That will be perfectly alright under cer-

tain circumstances! A special type of unmanned aerial vehicle (UAV) called Pouncer has been developed by Windhorse Aerospace that will be loaded with food, water, and medicine, and would fly independently to its pre-planned destination such as disaster-hit areas. It is a one-way flight drone whose pre-formed shell can be reused by the victims in disaster-hit areas for shelter and the frame can be burnt safely to cook food.

The company is planning to replace the drone structure with special materials like edible food components. That is, it will soon be getting an upgrade that allows some of its body parts to be consumed along with its cargo. Yes, it is possible but with some extra caution because after the coronavirus disease outbreak everything is not the same, especially in areas such as food hygiene and sanitation. Victims need to follow the WHO guidelines including hand hygiene before getting a bite or chunk of the edible drone.

The Pouncer will come in different sizes and capacities. The three-metre-wide drone was designed for hard-to-reach areas during disasters, natural calamities, and conflicts, delivering aid to the victims while saving time, money, and human lives. The drone will be launched from a catapult, balloon or aircraft depending on the area it is trying to reach. Frazer-Nash Consultancy and Windhorse Aerospace are in collaborative partnerships to develop a Pouncer made from a variety of materials to deliver humanitarian edible food aid to remote areas and war-torn locations. **EFY**



Fig. 4: Pouncer—edible drone (Credit: www.windhorse.aero)